Desk Tide

The tide is rising across your real desk. Your clutter is the coastline. Save the villages.

Gaming track | New Experience | Target: Best New Gaming Experience

AT A GLANCE

Track / division	Gaming (puzzle / strategy) / New Experience
Primary prize	Best New Gaming Experience ($100,000)
Fallback special award	Boldest Original Concept ($25,000), the strongest fit of all six ideas
Session length	One tide is about 8 minutes (three waves)
Room requirement	A desk with some objects on it; Quest 3 depth sensor strongly preferred
Core tech	Depth sensing to build a height map of the desk, a shallow-water simulation, hand tracking
Biggest risk	Real-world geometry capture: resolution, update rate and Quest 3S support
Solo feasibility	The hardest of the six ideas; needs a solid fallback mode

The one-line pitch

A miniature flood-defence puzzle game played on your actual desk. Water rises across the tabletop and flows around your real mug, keyboard and books, which become islands and cliffs. Pinch to place dams, drag a finger to dig channels, and move your real objects to reshape the land.

SECTION 01

The idea

Desk Tide shrinks a coastal world onto your desk. Tiny villages appear on the high ground around your real objects: on top of a book, beside your keyboard, on the base of your monitor. Then the tide comes in. Water spills across the tabletop, flows around the real obstacles and pools in the low spots. Your job is to keep the villages dry.

You defend them with your hands: pinch to drop sandbag dams, drag a fingertip across the desk to carve drainage channels, and pinch-rotate small pumps. You can also physically move real objects. Slide your mug into the gap and it becomes a sea wall. Every desk is a different map, and rearranging your desk rearranges the level.

Why it is the boldest idea

The Boldest Original Concept award asks for 'the idea that couldn't exist anywhere else' and whether the experience 'would fall apart on any other platform'. In Desk Tide the level is literally made from your real room. There is no version of it that works on a flat screen or in full VR.

SECTION 02

How a tide plays out

Phase	What happens	Player actions
Survey (1 min)	The desk is scanned. Villages appear on high ground. Faint arrows show where the water will enter (often from 'springs' at the base of real objects).	Look around, move real objects if you want, place the first dams.
Wave 1 (2 min)	A gentle rise. Water finds the low paths.	Place dams, dig channels.
Wave 2 (2 min)	A faster rise from a second direction.	Adapt; redirect water into 'safe basins'.
Wave 3 (2 min)	The high tide with a surge.	Last-second saves; pumps.
Result (1 min)	Saved villagers walk to the lighthouse on the desk corner.	Collect, then choose another tide or stop.

Tools

Tool	Gesture	Effect	Limit
Sandbag dam	Pinch from the tool tray, place on the desk, pinch-drag to extend.	Raises the terrain along a line.	Limited supply per tide
Channel	Drag an extended index finger along the desk.	Lowers the terrain along the path so water drains.	Takes time to dig
Pump	Pinch-place, then twist to set the direction.	Moves water from one cell area to another.	One or two per tide

Tool	Gesture	Effect	Limit
Real objects	Physically move a real object on your desk.	The terrain updates after a rescan (see technical notes).	Unlimited, but slow to register

SECTION 03

Why passthrough and the real room matter

- **The terrain is your desk.** Real objects are the islands, walls and cliffs. With passthrough off, the water would appear to flow around invisible things.
- **Every player gets a unique map.** Level generation is based on each player's own desk, so no two players see the same puzzle.
- **Physical rearrangement is a mechanic.** Moving a real mug to block a gap is a moment no other platform can offer.
- **Water hides behind real objects.** Using depth information, virtual water is correctly hidden behind the real objects it flows around.

SECTION 04

The first five minutes

Time	What happens
0:00	Passthrough. A shimmer passes over the desk as it is scanned. Real objects get a faint outline, then the outline turns into grassy tops and cliff edges.
0:30	Tiny villages pop up on the high ground with small sounds. A villager waves.
0:50	A single stream begins trickling from the base of the monitor toward a village. A glowing sandbag floats near the player's hand.
1:10	The player pinches the sandbag and drops it in the path. The water turns aside. The village cheers.
2:00	Wave 1 of the tutorial tide: two streams, one channel to dig.
4:30	Tide over. Saved villagers walk to the lighthouse on the corner of the desk, which stays there between sessions.

SECTION 05

Technical approach

Capturing the desk as a height map

The game needs a height map: a grid over the desk surface storing how high the real 'terrain' is at each point. There are three possible sources, in order of preference.

Source	How it works	Pros	Cons
Live depth sensing (Quest 3)	WebXR depth sensing gives a depth image of the real world. Sample it over the desk area and convert to heights.	Updates when you move objects; captures anything on the desk.	Low resolution; noisy on small or shiny objects; availability on Quest 3S must be verified.
Room mesh from Space Setup	The scene mesh captured during room setup, intersected with the desk plane.	Simple and stable.	Captured once during setup, so it does not update when objects move; too coarse for small items.
Player-placed blocks (fallback)	The player taps the top of each real object to create a block of that height.	Works everywhere, including on Quest 3S and on empty desks.	Less magical; more setup.

Recommendation

Build the player-placed block mode first, so the game works everywhere. Then add live depth scanning as an enhancement on Quest 3. Rescan the height map only during the Survey phase and when the player pinches the 'rescan' seashell on the tool tray, rather than continuously, to keep the simulation stable and the frame rate high.

Water simulation

- A shallow-water model on a grid of about 96 x 64 cells covering the desk (roughly 8 to 10 mm per cell).
- The 'virtual pipes' method: each cell exchanges water with its four neighbours based on the difference in surface height. It is simple, stable and well documented.
- Runs on the CPU at 30 steps per second with pre-allocated typed arrays (about 6,000 cells is cheap), or on the GPU if needed.
- Rendered as one transparent, gently animated surface mesh whose vertices follow the water height.

Level generation

- Villages are placed on local high points, with a minimum spacing.
- Water sources are placed at low points near the edge or at the base of tall objects.
- A quick simulation run with no player action checks the difficulty: the tide must threaten most villages but be saveable with the available tools.
- A daily tide uses the date as a random seed so difficulty is fair from day to day.

ECS design

System	Responsibility
DeskScanSystem	Builds the height map from depth, mesh or player-placed blocks.
LevelGenSystem	Places villages and sources; validates difficulty.
WaterSimSystem	Advances the shallow-water grid at a fixed rate; no allocation per step.
WaterRenderSystem	Updates the water surface mesh from the grid.
ToolSystem	Handles dam, channel and pump gestures and edits the terrain grid.
VillageSystem	Detects flooded villages; animates villagers; scores.
LighthouseSystem	The persistent reward on the desk corner; stored in IndexedDB.

SECTION 06

Progression and reasons to come back

System	Description
The Lighthouse town	Saved villagers settle around a lighthouse on your desk corner. It grows across sessions, with new houses, lights and boats. Visible growth that lives in your real space.
Daily Tide	A new seeded tide on your desk every day.
Seasons	Weekly themes: spring floods, storm season, frozen tide (ice blocks you can break).
Rearrange challenge	Occasional tides that can only be won by physically moving an object.

SECTION 07

Accessibility

- Relaxed mode: slower tides and unlimited tools.
- Pause at any time with a palm-up gesture.
- Colour-blind safe water and warning indicators (patterns, not only colour).
- One-handed play: every tool works with a single hand.

SECTION 08

Art direction and sound

Toy-like miniature style: soft pastel villages, clear teal water with foam edges, grassy tops on real objects. Sound: gentle waves, tiny villager voices, a satisfying thump when sandbags land. The small scale means small, cheap models that are easy to make or source under a suitable licence.

SECTION 09

Risks and mitigations

Risk	Severity	Mitigation
Depth data too coarse or noisy for small desk objects	High	Smooth and threshold the height map; ignore objects under about 2 cm; block-placement fallback.
Depth sensing unavailable on Quest 3S	High	Verify in week 1; block-placement mode is the 3S path.
Room mesh does not update when objects move	Medium	Use live depth for rescans; do not rely on the setup mesh.
Levels unsolvable or trivial on unusual desks	Medium	Simulated difficulty check at generation time; adjust tool supply.
Empty desk gives a boring map	Medium	Add virtual rocks and hills when the desk is flat.
Simulation and rendering cost	Medium	Fixed-rate simulation, modest grid, single water mesh.
Scope too big for one person	High	Ship the block mode and 3 waves first; depth scanning only if time allows.

Week-1 spike

1. Check that WebXR depth sensing runs in Quest Browser on your device and read depth values over the desk.
2. Build a height map from depth and render it as a debug mesh; see whether a mug and a book appear clearly.
3. Run a simple water grid on the height map at 30 steps per second and measure frame time.

SECTION 10

Seven-week build plan

Week	Dates	Goal	Exit criterion
Week 1	29 Sep - 4 Oct	Depth and height-map spike; water grid prototype	Water visibly flows around a real mug on device
Week 2	5 - 11 Oct	Block-placement fallback; tools: dam and channel	A full tide playable with blocks
Week 3	12 - 18 Oct	Level generation and difficulty check; villages and scoring	Three different desks give fair tides
Week 4	19 - 25 Oct	First five minutes; Lighthouse town; storage	A new player saves a village with no instructions
Week 5	26 Oct - 1 Nov	Depth rescans; pumps; Daily Tide; art pass	Content-complete
Week 6	2 - 8 Nov	Performance; playtests in 5+ rooms; accessibility options	72 fps; every tester's desk works
Week 7	9 - 15 Nov	Polish; record video; write submission	Video and text ready
Ship	16 - 18 Nov	Final checks; submit on 17 Nov	Submitted early

SECTION 11

Scope

Must ship	Nice to have	Cut
Block-placement terrain; water simulation; dams and channels; villages; 3-wave tide; Lighthouse	Live depth scanning; pumps; Daily Tide; seasons	Physics for real objects moving mid-tide; multiplayer; weather effects beyond water

SECTION 12

Demo video plan (under 3 minutes)

Time	Beat	Shot
0:00 - 0:10	Hook	Water floods across a real desk and splits around a real coffee mug; a tiny village on a book is threatened.
0:10 - 0:40	The premise	The scan turning real objects into islands.
0:40 - 1:40	Play	Real-time tide: dams, channels, a last-second save.
1:40 - 2:10	The magic moment	The player slides a real mug into a gap and the water turns aside.
2:10 - 2:50	Every desk is a level	Montage of different desks and maps; the Lighthouse town growing.

SECTION 13

Submission draft

Name	Desk Tide
Tagline (under 140 characters)	The tide is rising across your real desk. Your clutter is the coastline. Pinch, dig and rearrange to save the villages.
Track / division	Gaming / New Experience
Target launch	Web release Q1 2027

Description (first draft)

What if your messy desk were a coastline? Desk Tide scans the objects on your real desk and turns them into islands, cliffs and valleys for a tiny world of villages. Then the tide comes in.

Water flows across the tabletop, around your real mug and keyboard, and pools in the low spots. Pinch to drop sandbag dams, drag your finger to dig channels, and physically slide real objects into the gaps to build sea walls. Every desk is a different map, and rearranging your desk rearranges the puzzle.

We built Desk Tide with the Immersive Web SDK, turning Quest 3 depth data into a height map that drives a shallow-water simulation. On devices without depth, players tap real objects to mark them. Saved villagers settle into a lighthouse town on the corner of your desk that grows with every tide.

SECTION 14

How it maps to the judging criteria

Criterion (25% each)	Evidence in Desk Tide
Innovation and Creativity	The real room is the level; rearranging real objects is a mechanic.
Experience Design	Short tides, clear goals, physical interactions, passthrough essential.
Technical Implementation	Depth-based terrain, water simulation, occlusion. Also the riskiest area.
Polish and Presentation	A charming miniature look and an unforgettable hook shot.

Hard requirement and design guidelines

Test	What the rules ask	How this idea passes
Hands-first (hard requirement)	Complete the entire experience without ever pairing a controller.	Pinch-place, finger-drag and moving real objects by hand; no controllers.
Airplane seat test	Every interaction works within a two-foot radius, seated.	Everything happens on the desk surface in front of the player.

Test	What the rules ask	How this idea passes
One bus stop test	Fast cold start, clean pause/resume, a satisfying moment in 10 minutes or less.	A tide is about 8 minutes; the game pauses with a palm-up gesture or when the headset is removed.
Take it away test	Remove any third-party service and a project still stands.	No third-party services.

Competition rules that shape this plan

- **One prize per entry.** Rule 7 limits each entry to a single prize, so this brief aims at one track prize and names one special award as a fallback rather than trying to stack awards.
- **Submission:** a public HTTPS link (GitHub Pages, Vercel or similar) that stays live until the winner announcement (about 11 Dec 2026), a public YouTube/Vimeo video under 3 minutes, and the Devpost form (name, 140-character tagline, track, division, description of about 500 words, hand-interaction notes, target launch date).
- **No changes after the deadline.** Judging uses the build as it stood at 18 Nov 2026, 12:00 PT. Plan to submit on 17 Nov and freeze the hosted build.
- **Content rules:** English (or English subtitles), no identifiable real people other than entrants, no brand names or logos, all music, art and fonts properly licensed or original.
- **Video rules:** real footage from a Quest or the XR simulator; do not lean on AI-generated video to carry the pitch.
- **Performance floor:** judges expect at least 60 fps on Quest hardware. Target 72 fps.
- **'Holds up in rooms the developer never tested'** is the key technical criterion for this idea. The block-placement fallback is not optional.

SECTION 15

Honest weaknesses

- The technical risk is the highest of the six ideas; the magic depends on depth data quality.
- Quest 3S players may only get the fallback mode, which is less impressive.
- Messy real-world input makes level quality unpredictable.
- The water simulation, level generation and scanning together are a lot for one person in seven weeks.

SECTION 16

Scorecard and verdict

Dimension	Score / 10	Comment
Originality	10	Truly only possible in mixed reality.
Only possible in MR	10	The level is your desk.
Works in untested rooms	5	Depends on depth quality; the fallback saves it.
Solo feasibility in 7 weeks	5	Three hard systems at once.
Prize ceiling	$100k	Top Gaming prize; the best fit for Boldest Original Concept.

Verdict

The highest ceiling and the highest risk. Only choose it if the week-1 depth spike clearly works and you are comfortable shipping the block-placement mode as the reliable base.